

# Disproof of conjecture 00000008848

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## Verdict: FALSE

We disprove conjecture 00000008848 as stated. The conjecture has two clauses, and each fails:

- Clause (a), “minimal and maximal fixed points of order-preserving maps always exist”, fails already on the totally ordered set  $\mathbb{Z}$ : the shift  $f(x) = x + 1$  is order-preserving and has no fixed point at all, and the identity map is order-preserving with fixed-point set  $\mathbb{Z}$ , which has neither a least nor a greatest element.
- Clause (b), “monotone iterations of the sub/supersolution method approximate them in countably many steps”, fails even on a complete lattice: on the complete lattice  $\omega_1 + 1$  the map  $f(\alpha) = \alpha + 1$  for  $\alpha < \omega_1$ ,  $f(\omega_1) = \omega_1$ , is order-preserving with unique fixed point  $\omega_1$ , and the monotone iteration from the subsolution 0 reaches it only at stage  $\omega_1$ , i.e. in uncountably many steps.

The conjunction is therefore false under every reading. The honest caveat is this: the file specifies *no poset*, so a charitable reading that adds “complete lattice” makes clause (a) the Knaster–Tarski theorem and puts the  $\mathbb{Z}$  witnesses out of scope; under that reading the disproof of the conjunction rests on clause (b). We state this explicitly in Section 3 and again in the formalisation and reproducibility sections.

## 1 The conjecture and the hypotheses it omits

The conjecture, quoted verbatim from `conjectures/00000008848.md`:

**Definition:** Fixed points of order-preserving maps. **Conjecture:** Minimal and maximal fixed points of order-preserving maps always exist; and monotone iterations of the sub/supersolution method approximate them in countably many steps. (order-preserving sub/supersolution iteration)

The original bilingual statement (English and Chinese) is reproduced in full in `README.md`.

The statement is a template with several slots left unfilled.

1. **No poset is specified.** It says only “order-preserving maps”, with no hypothesis that the underlying ordered set is a complete lattice, a complete partial order, a lattice with a bottom, or even bounded. In particular  $\mathbb{Z}$  with its usual order is an admissible domain, and so is an arbitrary ordered set.

2. **No sub/supersolution hypothesis is stated** for the first clause. Clause (a) therefore asserts unconditional existence of a least and a greatest fixed point for every order-preserving self-map of every ordered set.
3. **No continuity hypothesis is stated** for the second clause — no sup-continuity, no Scott continuity, no preservation of directed suprema, no compactness. Without such a hypothesis, even on a complete lattice, transfinite monotone iteration may require uncountably many steps.

We now give the witnesses, first for clause (a) under the literal reading, then for clause (b) on a complete lattice. Throughout, “order-preserving” means *monotone*:  $a \leq b \Rightarrow g(a) \leq g(b)$ .

## 2 Clause (a) fails on ordered sets

### 2.1 The shift map on $\mathbb{Z}$ : no fixed point at all

**Definition 1.** Let  $f : \mathbb{Z} \rightarrow \mathbb{Z}$  be the shift  $f(x) = x + 1$ .

**Lemma 2.**  *$f$  is order-preserving.*

*Proof.* If  $a \leq b$  in  $\mathbb{Z}$  then  $a + 1 \leq b + 1$ , i.e.  $f(a) \leq f(b)$ . □

**Lemma 3.**  *$f$  has no fixed point:  $f(x) \neq x$  for every  $x \in \mathbb{Z}$ .*

*Proof.*  $f(x) = x$  would say  $x + 1 = x$ , whence  $1 = 0$ , a contradiction. □

**Corollary 4.** *The order-preserving map  $f$  on the ordered set  $\mathbb{Z}$  has neither a least nor a greatest fixed point; it has no fixed point whatsoever.*

Thus clause (a), read literally as a statement about all order-preserving maps of ordered sets, is false. Its failure is not merely a degenerate emptiness accident: the next subsection exhibits an order-preserving map whose fixed-point set is as large as possible yet still has no extremal element.

### 2.2 The identity map on $\mathbb{Z}$ : fixed points without extrema (non-vacuous)

**Theorem 5.** *The identity map  $\text{id} : \mathbb{Z} \rightarrow \mathbb{Z}$  is order-preserving, every point of  $\mathbb{Z}$  is a fixed point, but the fixed-point set  $\mathbb{Z}$  has neither a least nor a greatest element. Consequently there is no minimal fixed point and no maximal fixed point.*

*Proof.* Order-preservation is immediate:  $a \leq b \Rightarrow \text{id}(a) = a \leq b = \text{id}(b)$ . Every point is fixed because  $\text{id}(y) = y$  for all  $y$ .

Suppose for contradiction that  $m \in \mathbb{Z}$  were a least fixed point. Since every  $y$  is fixed, in particular  $y = m - 1$  is fixed, so minimality gives  $m \leq m - 1$ , which is false. Hence there is no least fixed point.

Similarly, if  $M$  were a greatest fixed point, then  $y = M + 1$  is fixed, so  $M + 1 \leq M$ , which is false. Hence there is no greatest fixed point either. □

*Remark 6.* Theorem 5 is the *primary* witness for clause (a) because it is non-vacuous: the map genuinely has fixed points (all of them), so the failure cannot be dismissed by saying the hypothesis “has a fixed point” is unmet. It shows that even the repaired statement

every order-preserving self-map of an ordered set that has at least one fixed point has a least fixed point,

is false, with  $\text{id}$  on  $\mathbb{Z}$  as witness.

**Corollary 7.** *Clause (a) is false as stated. Both the literal reading (no fixed point assumed; witness  $f(x) = x+1$  on  $\mathbb{Z}$ ) and the non-vacuous reading (a fixed point assumed; witness  $\text{id}$  on  $\mathbb{Z}$ ) are refuted.*

### 3 The reading caveat: clause (a) on a complete lattice

The refutation of clause (a) above uses only the fact that the file specifies no completeness. If a reviewer *charitably* supplies the hypothesis “complete lattice”, clause (a) becomes a true theorem, and the  $\mathbb{Z}$  witnesses fall outside its scope.

**Theorem 8** (Knaster–Tarski, for reference). *Let  $L$  be a complete lattice and  $g : L \rightarrow L$  order-preserving. Then  $g$  has a least fixed point, namely  $\inf\{x \in L : g(x) \leq x\}$ , and a greatest fixed point, namely  $\sup\{x \in L : g(x) \geq x\}$ ; indeed the fixed points of  $g$  form a complete lattice.*

*Proof sketch.* Let  $Q = \{x : g(x) \leq x\}$  be the set of post-fixed points. It is nonempty ( $\top \in Q$ ), so  $q = \inf Q$  exists; since  $g$  is monotone and every  $x \in Q$  satisfies  $q \leq x$ , one has  $g(q) \leq g(x) \leq x$  for all  $x \in Q$ , hence  $g(q) \leq q$ , so  $g(q) \in Q$  and  $q \leq g(q)$ ; therefore  $g(q) = q$  and  $q$  is the least fixed point. The greatest fixed point is dual, using  $\{x : g(x) \geq x\}$  and its supremum.  $\square$

*Remark 9.* Under this reading the witnesses  $f$  and  $\text{id}$  on  $\mathbb{Z}$  are out of scope because  $\mathbb{Z}$  is not a complete lattice: for instance the subset  $\mathbb{Z} \subseteq \mathbb{Z}$  has no supremum in  $\mathbb{Z}$ . We therefore do not claim that clause (a) is false on complete lattices. The next section shows that the *conjunction* is nevertheless false under the complete-lattice reading, because clause (b) fails there.

### 4 Clause (b) fails even on a complete lattice: $\omega_1 + 1$

We now exhibit a complete lattice and an order-preserving map for which the monotone sub/supersolution iteration converges only after uncountably many steps. This shows that “countably many steps” in the conjecture cannot hold without a continuity hypothesis the file does not state.

Let  $\omega_1$  denote the first uncountable ordinal and let

$$L = \omega_1 + 1 = \{\alpha : \alpha \leq \omega_1\},$$

ordered by  $\in$  (equivalently, by  $\leq$ ).

**Lemma 10.**  *$L$  is a complete lattice. Its least element is 0 and its greatest element is  $\omega_1$ .*

*Proof.* The order is that of the ordinals. For a nonempty subset  $S \subseteq L$  the infimum is  $\min S$ , which exists because the ordinals are well-ordered; for  $S = \emptyset$  the infimum is the top element  $\omega_1$ . For the supremum,  $S = \emptyset$  gives 0. If  $S$  is nonempty and bounded in  $L$  by some  $\alpha < \omega_1$ , then  $S \subseteq \alpha$  is a set of ordinals below a countable ordinal, hence  $S$  is countable, and its supremum  $\sup S = \bigcup S$  is an ordinal that is a union of countably many countable ordinals, hence countable, so  $\sup S < \omega_1$  lies

in  $L$ . If  $S$  is unbounded below  $\omega_1$ , then  $\sup S = \omega_1 \in L$ . Hence every subset of  $L$  has a supremum, and  $L$  is a complete lattice.  $\square$

**Definition 11.** Define  $g : L \rightarrow L$  by

$$g(\alpha) = \begin{cases} \alpha + 1, & \alpha < \omega_1, \\ \omega_1, & \alpha = \omega_1. \end{cases}$$

**Lemma 12.**  $g$  is order-preserving.

*Proof.* Let  $\alpha \leq \beta$  in  $L$ . If  $\beta < \omega_1$  then also  $\alpha < \omega_1$ , so  $g(\alpha) = \alpha + 1 \leq \beta + 1 = g(\beta)$ . If  $\beta = \omega_1$  then  $g(\beta) = \omega_1$  is the greatest element of  $L$ , so  $g(\alpha) \leq \omega_1 = g(\beta)$  automatically.  $\square$

**Lemma 13.**  $g$  has exactly one fixed point, namely  $\omega_1$ . Hence  $\omega_1$  is both the least and the greatest fixed point of  $g$ .

*Proof.* For  $\alpha < \omega_1$  we have  $g(\alpha) = \alpha + 1 > \alpha$ , so  $\alpha$  is not fixed. And  $g(\omega_1) = \omega_1$ . Thus the fixed-point set is  $\{\omega_1\}$ , a singleton.  $\square$

The conjecture's second clause concerns the monotone *sub/supersolution iteration*. Recall the standard transfinite iteration from below: given a subsolution  $u$  (a point with  $g(u) \geq u$ ), define  $\{u_\alpha\}$  by

$$u_0 = u, \quad u_{\alpha+1} = g(u_\alpha), \quad u_\lambda = \sup_{\beta < \lambda} u_\beta \quad (\lambda \text{ a limit ordinal}),$$

continuing for as long as the suprema exist; on a complete lattice they always do. Since  $g$  is order-preserving, the sequence is increasing and each  $u_\alpha$  is again a subsolution, and its limit is the greatest fixed point below the starting value — in particular, starting at the least element yields the least fixed point.

Take the subsolution  $u_0 = 0$ : indeed  $g(0) = 1 \geq 0$ , and 0 is the least element of  $L$ .

**Lemma 14.** For every ordinal  $\alpha \leq \omega_1$  the transfinite iteration from 0 satisfies  $u_\alpha = \alpha$ .

*Proof.* By transfinite induction on  $\alpha \leq \omega_1$ .

*Zero:*  $u_0 = 0$  by definition.

*Successor:* let  $\alpha + 1 \leq \omega_1$ , so  $\alpha < \omega_1$ . By the induction hypothesis  $u_\alpha = \alpha < \omega_1$ , hence  $u_{\alpha+1} = g(u_\alpha) = g(\alpha) = \alpha + 1$ .

*Limit:* let  $\lambda \leq \omega_1$  be a limit ordinal. By the induction hypothesis  $u_\beta = \beta$  for all  $\beta < \lambda$ , so  $u_\lambda = \sup_{\beta < \lambda} \beta = \lambda$ , the last equality holding because the ordinals below a limit ordinal are cofinal in it. (For  $\lambda = \omega_1$  this reads  $\sup_{\beta < \omega_1} \beta = \omega_1$ .)  $\square$

**Theorem 15.** On the complete lattice  $L = \omega_1 + 1$ , the order-preserving map  $g$  of Definition 11 has the unique fixed point  $\omega_1$  (Lemma 13), but the monotone iteration from the subsolution 0 satisfies  $u_\alpha = \alpha$  for all  $\alpha \leq \omega_1$  (Lemma 14). Consequently:

1. for every countable ordinal  $\alpha < \omega_1$  we have  $u_\alpha = \alpha < \omega_1$ , so the iteration has not reached the least fixed point at any countable stage;

2. the first stage at which it reaches the fixed point is  $\alpha = \omega_1$ , the first uncountable ordinal.

Hence the sub/supersolution iteration requires uncountably many steps, not countably many. Clause (b) is false on a complete lattice.

*Proof.* Immediate from Lemmas 13 and 14:  $u_\alpha = \alpha$  equals  $\omega_1$  exactly when  $\alpha = \omega_1$ . Since every  $\alpha < \omega_1$  is countable and  $\omega_1$  is uncountable, convergence occurs only at an uncountable stage.  $\square$

*Remark 16* (Why continuity is exactly what is missing). The map  $g$  is not sup-continuous. On the directed set  $D = \omega = \{0, 1, 2, \dots\} \subseteq L$  we have  $\sup D = \omega$  and

$$g(\sup D) = g(\omega) = \omega + 1 \neq \omega = \sup_{n \in D} g(n) = \sup_{n < \omega} (n + 1).$$

This failure is precisely what prevents convergence at stage  $\omega$ : at that stage  $u_\omega = \omega$  while  $g(u_\omega) = \omega + 1 > u_\omega$ , so the iteration must continue. Had the file assumed that  $g$  preserves directed suprema (Scott continuity), the iteration would indeed converge after countably many steps,  $u_\omega$  being the least fixed point. The conjecture states no such hypothesis, and Theorem 15 shows the conclusion fails without it.

*Remark 17* (Reading of “countably many steps”). Under either reading — a sequence indexed by the natural numbers, or a transfinite iteration indexed by the countable ordinals — the conclusion fails: the iteration is not constant at any countable stage, and its first fixed value occurs at the uncountable ordinal  $\omega_1$ . The singleton fixed-point set makes the example as non-degenerate as possible on the fixed-point side: minimal and maximal fixed points coincide, so there is no ambiguity about which one is being approximated.

## 5 Conclusion

Clause (a) is false on ordered sets, and clause (b) is false even on a complete lattice. Therefore conjecture 00000008848 is false as stated, under the literal reading and under the charitable complete-lattice reading alike. For the record, the logical structure of the disproof is:

| Reading                             | Clause (a)                         | Clause (b)                  |
|-------------------------------------|------------------------------------|-----------------------------|
| Literal (no completeness; as filed) | false (§3, $\mathbb{Z}$ witnesses) | false (§5, $\omega_1 + 1$ ) |
| Charitable (add “complete lattice”) | true (Knaster–Tarski, §4)          | false (§5, $\omega_1 + 1$ ) |

In both rows the conjunction is false. Under the charitable reading the disproof rests on clause (b), exactly as stated in the introduction.

## 6 Formalisation

Clause (a) is formalised in the accompanying Lean 4 project under `lean4/`, using core Lean only (`import Std`, no `Mathlib`, no `sorry`); `lake build` succeeds and `lake env lean Check.lean` reports the axioms of every theorem with no `sorryAx`. The formal statements include, for  $\mathbb{Z}$ ,

```
theorem f_mono : ∀ a b : Int, a ≤ b → f a ≤ f b
theorem f_fixfree : ∀ x : Int, f x ≠ x
theorem refutes_least_fixed_point :
  ¬ (∀ (α : Type) [LE α] (g : α → α),
```

```

      (∀ a b, a ≤ b → g a ≤ g b) →
      ∃ m, g m = m ∧ ∀ y, g y = y → m ≤ y)
theorem id_no_least_fixed :
  ¬ ∃ m : Int, IsFixed m ∧ ∀ y, IsFixed y → m ≤ y
theorem refutes_least_fixed_point_with_fixpoint : (the non-vacuous form)

```

The last two formalise Theorem 5 and the non-vacuous refutation; `refutes_least_fixed_point` formalises the shift-map witness of Corollary 4; the greatest-fixed-point halves are formalised in the same file.

Clause (b) is *not* formalised: core Lean (and core Lean’s standard library) has no ordinals or cardinals, and Mathlib, which does, is not available in this project. The  $\omega_1 + 1$  argument is therefore carried out in this document, and illustrated (not computed) in `reproduce.py`. This limitation is stated in `lean4/README.md`.

## 7 Reproducibility

`reproduce.py` (Python 3 standard library only) verifies:

1. that  $f(x) = x + 1$  on  $\mathbb{Z}$  has no fixed point, by checking  $x + 1 \neq x$  on a large integer window and by the exact algebraic identity  $x + 1 = x \iff 1 = 0$ ;
2. that `id` on  $\mathbb{Z}$  has every point fixed and no least or greatest fixed point: for any claimed least element  $m$  it exhibits the fixed point  $m - 1 < m$ , and for any claimed greatest element  $M$  it exhibits the fixed point  $M + 1 > M$ , reporting the contradiction;
3. that the transfinite iteration for  $\omega_1 + 1$  needs more than countably many steps (Thm. 5 in the script): it *simulates* the first finitely many stages on a finite initial segment, showing  $u_n = n$  for all simulated  $n$ , and then records the standard transfinite argument of Lemma 14 and Theorem 15 as the actual proof, clearly labelled as *argument*, not computation.

It prints `PASS` and exits 0 exactly when every check holds, and exits non-zero otherwise.

## 8 Status against the submission rules

Rule 3 of the competition requires each submission to contain the LaTeX source code, a PDF document, and a Lean 4 project.

- **LaTeX source** — `main.tex`, a standalone article using only `geometry`, `amsmath`, `amssymb`, `amsthm`, `array`, `parskip`, and `booktabs`; it compiles with `tectonic`.
- **PDF document** — `build/main.pdf`, produced by `tectonic --outdir build main.tex`. (Plain `tectonic main.tex` also compiles, writing `./main.pdf`.)
- **Lean 4 project** — `lean4/` with `lean-toolchain` (`leanprover/lean4:v4.33.1`), `lakefile.toml` (project `tlmc8848`), `Main.lean`, `Check.lean`, and a `README.md`. It formalises clause (a) completely and audits its axioms; clause (b) is argued here because core Lean has no ordinals.